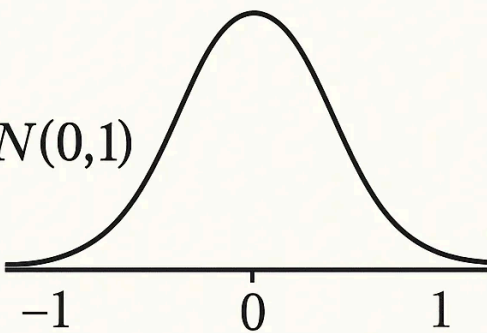


Central Limit Theorem

The CLT states that the distribution of sample means approximates a normal distribution (a bell curve) as the sample size becomes larger, regardless of the population's underlying distribution shape.

CENTRAL LIMIT THEOREM

The Central Limit Theorem states that when independent random variables are added, their properly normalized sum tends toward a normal distribution.

$$\begin{array}{c} X_1 \\ \downarrow \\ X_2 \\ \downarrow \\ \vdots \\ X_n \end{array} \quad Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \rightarrow N(0,1)$$


For example: get the dataset as skew as you want, then take a few samples, always the same numbers, and look at the average. and do this many times and plot all these average, you get the normal distribution.